

What is CV?

↳ is a subset of AI where we provide ability to machine to process visual info like image, video etc.

Applications of CV:

① Image classification



② Object Recognition



③ Image segmentation

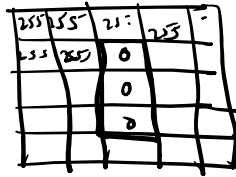
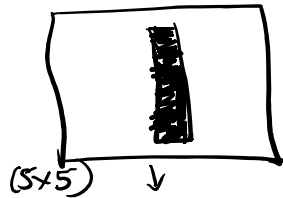


CNN → convolutional Neural Network is an algorithm introduced to specifically work on visual data.

What is digital image? → is a collection of pixels (number which represents colors → 0 to 255) (2D representation)

① Black white image (Grey Scale)

0 → black  
255 → white



② RGB scale image

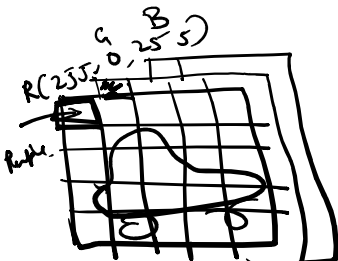
0 to 255 for R, G, B

(0, 0, 0) → Black image

(255, 0, 0) → Red image

(0, 255, 0) → Green

(0, 0, 255) → Blue





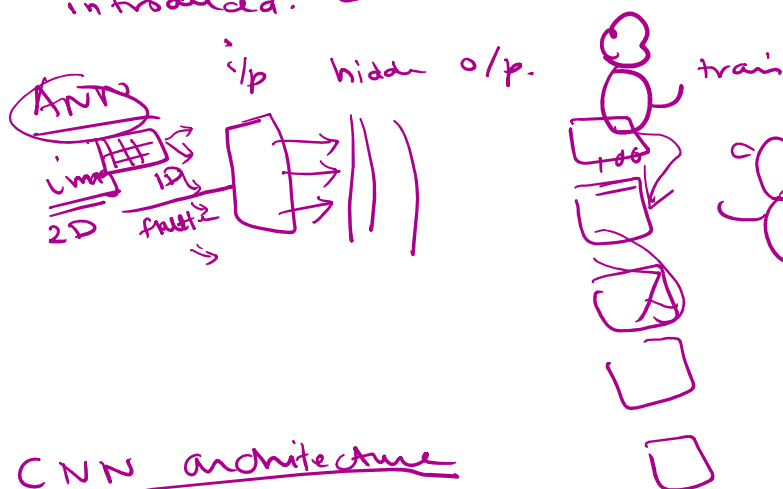
2 yr

CNN

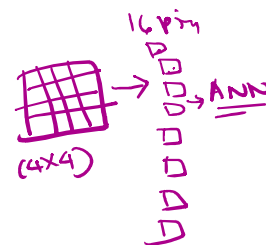
Occipital lobe in brain has different layers. Each layer is responsible for different task.

One layer  $\rightarrow$  identify shape  
 another layer  $\rightarrow$  identify edges/corner.  
 another layer  $\rightarrow$  identify objects within an image.

Using this working on Occipital lobe, CNN was introduced.



ANN Disadv  
visual image process

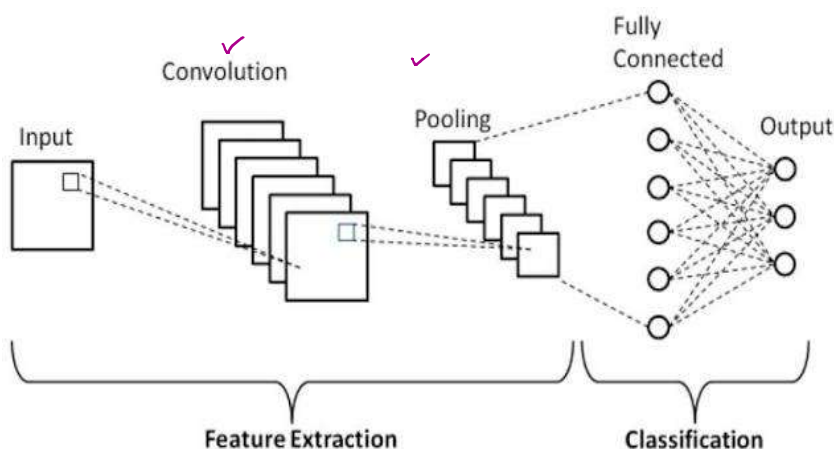


① Lot of training data has to be provided. Fails to identify different orientations things

② ANN requiring 2D images to be flatten to 1D pixel repr. & during training it fails to identify relationships between independent pixel.

③ Computational time & cost is higher with ANN.

CNN architecture

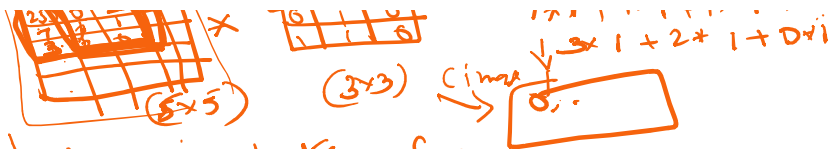


① Convolution layer  $\rightarrow$  where we extract imp feature from an image by performing

convolution operation on filters (Kernels).

$$\begin{array}{|c|c|c|c|} \hline 2 & 0 & 2 & 1 \\ \hline 2 & 3 & 1 & 1 \\ \hline 3 & 1 & 1 & 0 \\ \hline \end{array} \times \begin{array}{|c|c|c|} \hline 1 & 0 & 1 \\ \hline 0 & 1 & 0 \\ \hline 1 & 1 & 0 \\ \hline \end{array} = \begin{array}{l} (2 \times 5 \times 1 + 2 \times 3 \times 1 + 2 \times 1) + \\ 7 \times 1 + 1 \times 1 + 1 \times 1 + \\ 3 \times 1 + 2 \times 1 + 0 \times 1 \end{array}$$

(3x3) (image)



Stride → step size taken for matrix multiplication

Padding → adding extra layer to the image to avoid shrinking of input image when its multiplied by filter.

|                                                                                                                                                             |              |                     |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------|---|---|---|---|---|---|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| <div><div>✓</div><table><tr><td>3</td><td>5</td><td>1</td></tr><tr><td>3</td><td>6</td><td>1</td></tr><tr><td>4</td><td>7</td><td>9</td></tr></table></div> | 3            | 5                   | 1 | 3 | 6 | 1 | 4 | 7 | 9 | <table><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>3</td><td>5</td><td>1</td><td>0</td></tr><tr><td>0</td><td>3</td><td>6</td><td>1</td><td>0</td></tr><tr><td>0</td><td>4</td><td>7</td><td>9</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table> | 0 | 0 | 0 | 0 | 0 | 0 | 3 | 5 | 1 | 0 | 0 | 3 | 6 | 1 | 0 | 0 | 4 | 7 | 9 | 0 | 0 | 0 | 0 | 0 | 0 | <table><tr><td>3</td><td>3</td><td>5</td><td>1</td><td>1</td></tr><tr><td>3</td><td>3</td><td>5</td><td>1</td><td>1</td></tr><tr><td>3</td><td>3</td><td>6</td><td>1</td><td>1</td></tr><tr><td>4</td><td>4</td><td>7</td><td>9</td><td>9</td></tr><tr><td>4</td><td>4</td><td>7</td><td>9</td><td>9</td></tr></table> | 3 | 3 | 5 | 1 | 1 | 3 | 3 | 5 | 1 | 1 | 3 | 3 | 6 | 1 | 1 | 4 | 4 | 7 | 9 | 9 | 4 | 4 | 7 | 9 | 9 |
| 3                                                                                                                                                           | 5            | 1                   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3                                                                                                                                                           | 6            | 1                   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4                                                                                                                                                           | 7            | 9                   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                           | 0            | 0                   | 0 | 0 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                           | 3            | 5                   | 1 | 0 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                           | 3            | 6                   | 1 | 0 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                           | 4            | 7                   | 9 | 0 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                           | 0            | 0                   | 0 | 0 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3                                                                                                                                                           | 3            | 5                   | 1 | 1 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3                                                                                                                                                           | 3            | 5                   | 1 | 1 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3                                                                                                                                                           | 3            | 6                   | 1 | 1 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4                                                                                                                                                           | 4            | 7                   | 9 | 9 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4                                                                                                                                                           | 4            | 7                   | 9 | 9 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| No padding                                                                                                                                                  | Zero padding | Replication padding |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

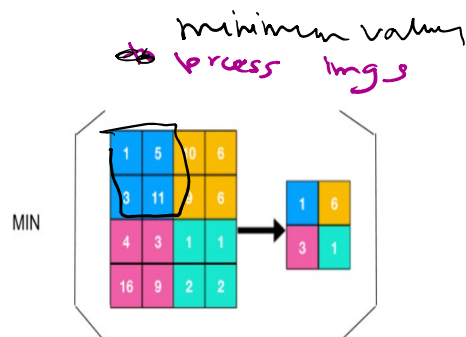
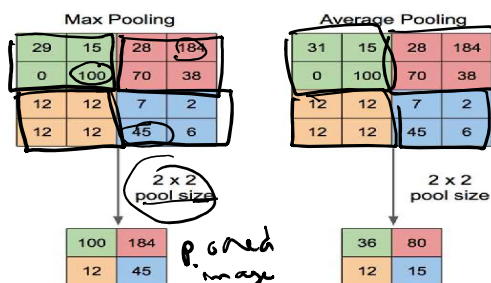
Pooling layer:

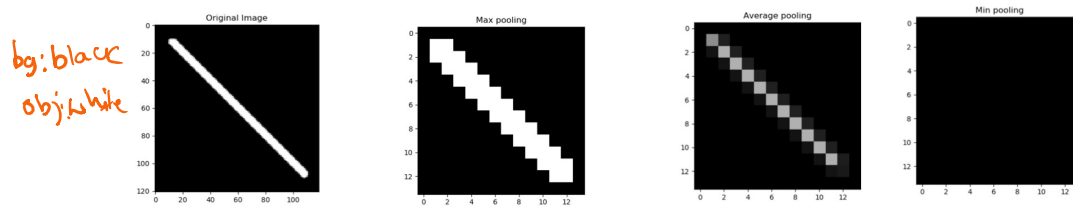
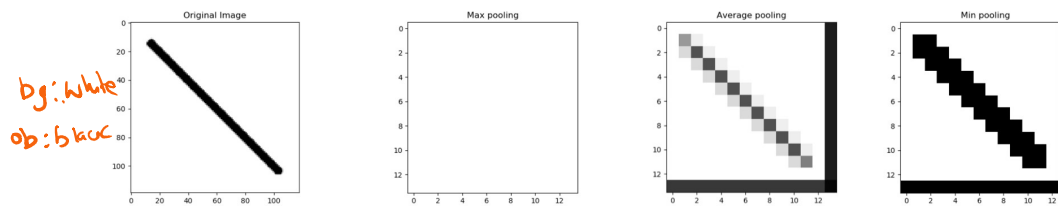
3x3x6

o/p of convolution layer i.e feature map is passed through pooling layer.

Here we down sample the image.

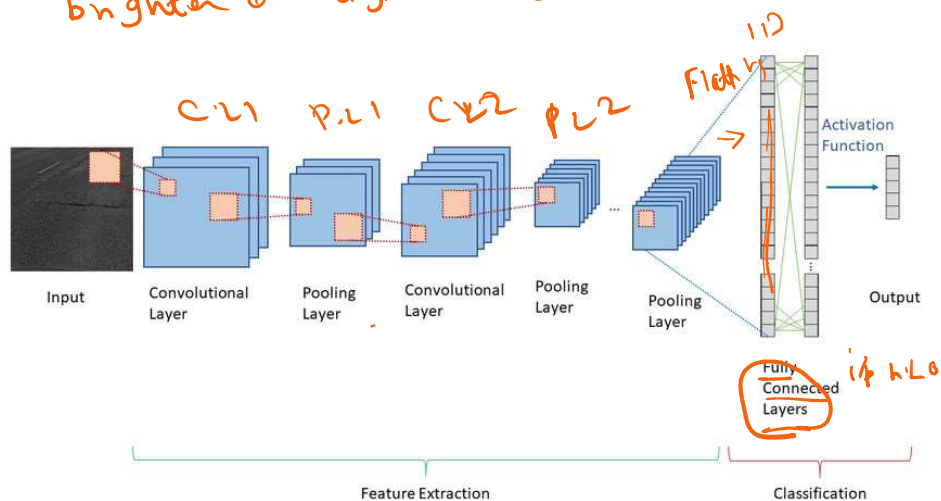
↑  
reduce size/dimension



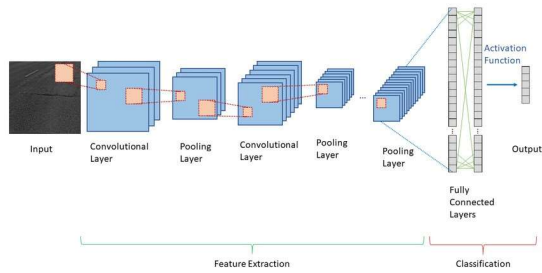


process imgs

- \* Min pools works better when object is darker & background is brighter.
- \* max pool works better when object is brighter & background is darker.

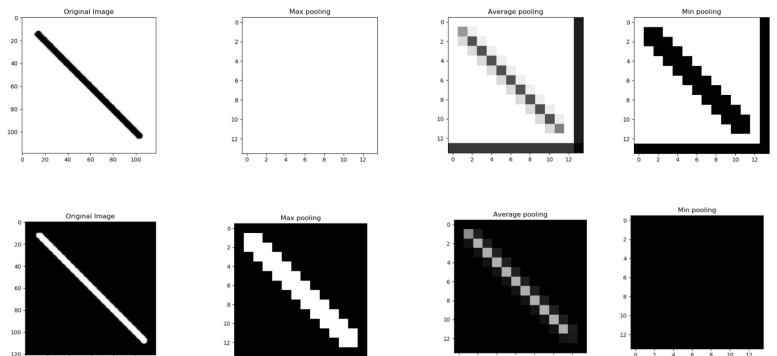
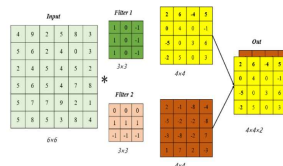


|                                                                                                                                      |              |                     |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
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|                                                     |              |                     |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
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| 3                                                                                                                                    | 5            | 1                   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3                                                                                                                                    | 6            | 1                   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4                                                                                                                                    | 7            | 9                   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                    | 0            | 0                   | 0 | 0 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                    | 3            | 5                   | 1 | 0 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                    | 3            | 6                   | 1 | 0 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                    | 4            | 7                   | 9 | 0 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                    | 0            | 0                   | 0 | 0 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3                                                                                                                                    | 3            | 5                   | 1 | 1 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3                                                                                                                                    | 3            | 5                   | 1 | 1 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3                                                                                                                                    | 3            | 6                   | 1 | 1 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4                                                                                                                                    | 4            | 7                   | 9 | 9 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4                                                                                                                                    | 4            | 7                   | 9 | 9 |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| No padding                                                                                                                           | Zero padding | Replication padding |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

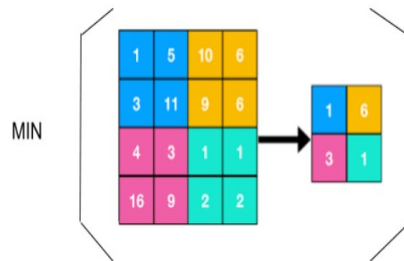


| Max Pooling     |     |    |     |
|-----------------|-----|----|-----|
| 29              | 15  | 28 | 184 |
| 0               | 100 | 70 | 38  |
| 12              | 12  | 7  | 2   |
| 12              | 12  | 45 | 6   |
| 2 x 2 pool size |     |    |     |
| 100             | 184 |    |     |
| 12              | 45  |    |     |

| Average Pooling |     |    |     |
|-----------------|-----|----|-----|
| 31              | 15  | 28 | 184 |
| 0               | 100 | 70 | 38  |
| 12              | 12  | 7  | 2   |
| 12              | 12  | 45 | 6   |
| 2 x 2 pool size |     |    |     |
| 36              | 80  |    |     |
| 12              | 15  |    |     |



to process imgs

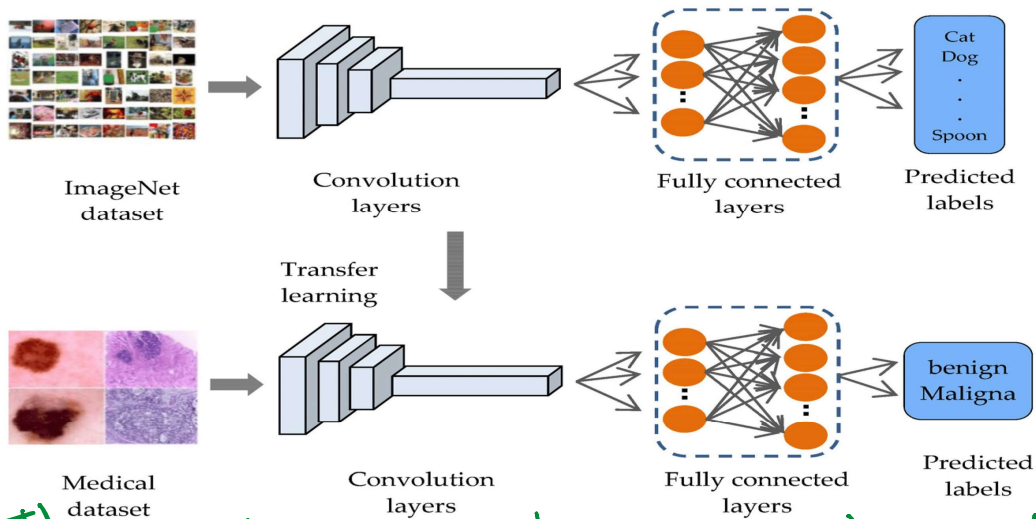


Transfer learning is a Machine learning technique where we take a model that has already been trained on a large dataset and reuse its learned knowledge for a new related task.

## Example

- \* Model trained to identify cars can be trained to identify trucks.
- \* Model trained to identify dogs, cats, flowers can be trained to recognise healthy vs diseased plant leaves.

Here, instead of training a new image classification from scratch, we can take this pre-trained model and train only the final layer to recognise.



The model already knows how to detect edges, shapes & patterns.

Pre-trained model

↓  
 Reuse learned features  
 ↓  
 Add new output layer  
 ↓  
 Train on new dataset  
 ↓  
 New model.

Advantages:

- \* Requires less training data
- \* Faster than training from scratch.
- \* Requires less computational power.

- \* require less computational power.
- \* Often provide better accuracy.
- \* Very useful when the new dataset is small.

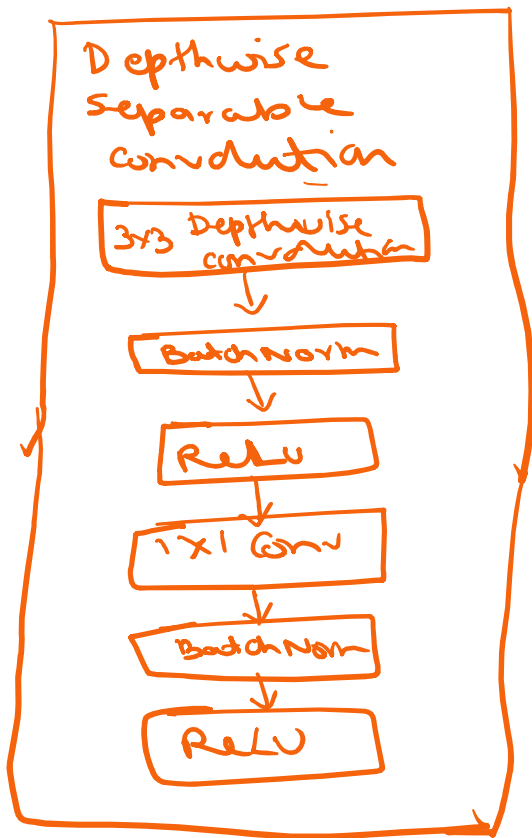
## Popular Pretrained Models

- \* VGG G16
- \* MobileNet
- \* EfficientNet
- \* Inception
- \* DoodleNet

## MobileNet

It is a lightweight CNN architecture designed mainly for mobile devices, embedded systems where computational resources are limited.

MobileNet uses Depthwise Separable Convolution. It significantly reduces computation, no. of parameters & memory usage.



Depthwise Convolution - applying one filter to each input channel separately

(1x1) Pointwise Convolution  
Combining info from different channels.

This makes Mobilenet more efficient.

Ex:-

I/p:- 1000 images of different plant diseases  
Training a CNN from scratch may require lot of data & computation.

Hence, we can use Mobilenet here

↓  
Reuse its learned features  
↓

Train it on your plant disease dataset

↓  
get a plant disease classifier.

## VGG16

VGG16 is a deep CNN developed by the Visual Geometry Group at the University of Oxford.

VGG16 has 16 learnable layers:-

- \* 13 Convolutional layers.
- \* 3 fully connected (dense) layers.

\* It is relatively slower than Mobilenet & not ideal for mobile devices.

\* VGG16 uses  $3 \times 3$  convolution filter with a stride of 1.

Example:-

Classification of cats vs dogs.

Use VGG16 trained on ImageNet

↓  
Since it ~~also~~ already understands  
edges, shapes, freeze convolution  
layer

↓  
Add a new classifier

↓  
Train on cat/dog images